Leveraging Large Language Models for Personalized Parkinson's Disease Treatment

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***Abstract*—Parkinson's Disease (PD) treatment is challenging due to symptom heterogeneity and the lack of a definitive cure. Lifelong medication requires personalized treatment plans developed by physicians, but such approaches are constrained by high costs and limited physician capacity. Although deep learning (DL) methods have been explored, they lack interpretability and are restricted to numerical data inputs. In this study, we propose a novel framework that leverages large language models (LLMs) to design personalized PD treatment strategies, integrating both patient information in natural language form and external textual knowledge sources (e.g., medical guidelines). To enhance effectiveness, we use Monte Carlo Tree Search (MCTS) to refine strategies and establish a robust medication recommendation dataset. To enhance reliability and interpretability, we incorporate Retrieval-Augmented Generation (RAG) and Chain-of-Thought (CoT) reasoning within the LLM system, ensuring that each proposed strategy is accompanied by step-by-step explanations and references to similar historical cases. Experimental evaluations using the Parkinson's Progression Marking Initiative (PPMI) dataset show that our method surpasses physician-prescribed treatments, achieving an average reduction of over 1.4 points in the revised unified Parkinson's disease rating scale part III (MDS-UPDRS-III) scores. Our method also outperforms the RL-method by 1.01 points on average. Furthermore, over 43% of patients achieve more than 2 point-reduction of MDS-UPDRS-III scores. A detailed case study highlights the flexibility of LLMs in dynamically adjusting medication plans for patients at different disease stages, highlighting its potential to advance personalized PD management in real-world settings.**

***Index Terms*—Parkinson's Disease (PD), LLM, treatment strategies, personalized prescription.**

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Data is available on-line at https://https://github.com/360ZMEM/PD-LLM.

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I. Introduction

PARKINSON'S Disease (PD) poses a major challenge that exacerbates public health pressures. According to the World Health Organization (WHO) [1], approximately 7 to 10 million individuals are currently living with Parkinson's Disease globally, the majority of whom are elderly. As the global aging population continues to rise, the number of PD patients will increase to between 12 and 17 million by 2040 [2].

Despite significant advances in medical science, PD remains an incurable neurodegenerative disorder. Traditional treatment strategies primarily aim to alleviate symptoms rather than halt or reverse the progression of the disease. Levodopa, the most commonly prescribed medication, can temporarily improve motor symptoms by replenishing dopamine levels in the brain. However, due to the complex drug-release mechanism of Levodopa, physicians often need to adjust dosages through trial and error, a process that is not only time-consuming but also prone to imprecision. Traditional medication regimens are heavily reliant on the expertise of neurologists, who are tasked with the continuous monitoring and individualized adjustment of drug therapies based on the patient's symptoms. This results in high treatment costs and frequent patient visits, further placing a strain on healthcare systems.

Moreover, the number of neurologists is insufficient to meet the growing demand for PD care [3], meaning that many patients are unable to receive timely and effective treatment. Neurodegenerative diseases, including Parkinson's Disease, are expected to pose significant challenges to public health in the coming decades, necessitating the development of more intelligent and effective management strategies to mitigate the growing burden on healthcare systems.

Recent advancements in deep learning (DL) have enabled the exploration of AI to assist in healthcare. While DL models based on clinical tabular data have shown promise in PD treatment, they face three key limitations: **effectiveness, reliability, and interpretability**. First, deep learning systems that rely solely on numerical data, while neglecting textual information, are inherently constrained in their ability to process PD comprehensively. Numerical data lacks semantic context (e.g., clinical interpretation of medications or timing) [4], and can result in partial information loss [5]. This reduces the clinical relevance of these models. Second, the reliability of AI systems directly impacts both patient safety and therapeutic efficacy [6]. These systems should be capable of accessing real-world medical

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knowledge and adopting evidence-based drug combinations. Third, while deep learning-based systems can generate treatment recommendations, their decision-making processes are often closed box, making it challenging for physicians and patients to understand the rationale behind suggested treatment plans [7]. Furthermore, physicians should retain the ability to intervene and provide feedback to further optimize the AI system's decision-making [8].

To address these limitations, we require a model capable of jointly processing both numerical and textual data in Parkinson's disease. Large language models (LLMs) offer a suitable architecture for this purpose. Prior methods, such as hybrid fusion networks combining tabular and text data, often require custom architectures and suffer from poor feature alignment [9] and overfitting on limited PD datasets [10]. Similarly, graph-based methods struggle to dynamically update structures to accommodate evolving datasets [11]. In comparison, LLMs integrate structured inputs (e.g., clinical scores) and unstructured text (e.g., medical guidelines) with minimal feature engineering [12]. By employing Retrieval-Augmented Generation (RAG), LLMs retrieve external information, such as medical guidelines, to enhance responses accuracy [13]. Additionally, Chain-of-Thought (CoT) prompting enables step-by-step reasoning, improving interpretability in complex clinical tasks [14]. Moreover, LLMs enable scalable knowledge expansion without retraining, making them well-suited for dynamic clinical contexts.

Motivated by these advantages, this paper designs and validates an LLM-based decision-making system for PD treatment. Specifically, we apply Monte Carlo Tree Search (MCTS) to identify optimal treatment strategies by simulating various treatment paths and their corresponding outcomes within a constrained search space. Following this, the LLM, acting as a "virtual physician", utilizes both numerical and textual patient data, alongside the medication dataset generated by MCTS, to deliver personalized treatment recommendations. To further enhance the system's capabilities, we integrate RAG and CoT reasoning techniques within the LLM system. This integration ensures that each treatment strategy is accompanied by detailed explanations and references to similar historical cases. Finally, we use the PPMI dataset, a real-world clinical database for PD, and incorporate medical guidelines to construct a local knowledge vector database that supports these processes.

To assess the performance of this LLM-based approach, we compare three different treatment strategies for PD: (1) clinician-prescribed medication plans, (2) Reinforcement learning (RL) methods trained solely on numerical data, and (3) our proposed PD-LLM framework, which integrates both numerical and textual information. This comparison enables us to assess the added value of external knowledge incorporation. The main contributions of this work are as follows:

- *Development of a Novel Decision-Making System:* We propose the first decision-making system for PD treatment based on LLMs, which incorporate natural language inputs, including medical guidelines and patient information records. This approach enhances the personalized treatment recommendations, improving the efficacy of PD management.
- *Integration of Advanced AI Techniques:* For **effectiveness**, we incorporate textual data and use numerical data in its raw form to maintain accuracy. We have developed a workflow that combines traditional data-driven methods with LLMs, using MCTS to identify optimal treatment plans and create a corresponding medication dataset. For **reliability**, we integrate RAG into the LLM framework, enabling the retrieval of relevant historical cases. To improve **interpretability**, the system applies CoT reasoning to provide treatment recommendations accompanied by detailed explanations, making decisions transparent. Additionally, this system can further engage in interactions with both physicians and patients, showcasing its potential for interactivity.
- *Exceptional Performance on Real-World Datasets:* Using the PPMI dataset, we conduct a comprehensive comparison of our LLM-based approach with traditional deep learning and physician methods. The results highlight the potential of LLMs in advancing PD treatment strategies.

II. Related Work

A. Data-Driven and Machine Learning-Based Parkinson's Disease Treatment

Current AI-powered approaches to PD management primarily focus on diagnostic or detection tasks, with comparatively less emphasis on treatment. Treatment strategies, by contrast, often focus on developing personalized, fine-grained medication policies for individual PD patients. We have summarized recent methods for Parkinson's treatment in Table I.

Watts et al. [15] proposed a Deep Reinforcement Learning (DRL)-based framework for optimizing medication dosage in Parkinson's Disease. However, this approach relies on assumed patient data and has not been validated on real-world datasets. Kim et al. [16] classified patients into 28 discrete states using decision trees, subsequently applying a Markov Decision Process (MDP) to derive the optimal medication combination for each state, excluding medication dosages. While the simplification of states and policies contributes to some degree of interpretability, it fails to provide personalized medication strategies for individual patients. Baucum et al. [17] integrated DRL with wearable device datasets to propose a personalized, dynamically adjustable medication dosage adjustment method. Experimental results demonstrated that the RL-based approach outperformed physicians in treatment efficacy. Riasi et al. [18] employed a recurrent neural network (RNN) model to predict medication dosages, thus overcoming the limitations associated with the curse of dimensionality in RL action spaces. However, regression-based methods are confined to replicating physician strategies and lack the capability to explore or identify more optimal treatment solutions.

While these studies highlight the potential of AI in PD treatment, they also expose several critical limitations. First, the non-linear nature of neural networks often leads to a lack of interpretability, making it difficult to clearly explain the decision-making process. Furthermore, effective PD treatment

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TABLE I
COMPARISON OF THIS WORK WITH EXISTING METHODS

Methods	Real-world data	Dosage data	Personalized treatment plans	Textual information	Knowledge base	Explainable treatment plans
Watts *et al.* [15]	✗	✗	✔	✗	✗	✗
Kim *et al.* [16]	✔	✗	✗	✗	✗	✔
Baucum *et al.* [17]	✔	✔	✔	✗	✗	✗
Riasi *et al.* [18]	✔	✔	✔	✗	✗	✗
Ours	✔	✔	✔	✔	✔	✔

necessitates the integration of patient history, medical guidelines, and physicians' diagnostic opinions, all of which are textual data that play an integral role in clinical treatment. Neglecting these elements may result in suboptimal treatment outcomes. In conclusion, despite existing efforts to develop AI-assisted strategies for PD management, ensuring the effectiveness, reliability, and interpretability of these systems still remains a significant challenge.

B. Large Language Model Agent

Large language models (LLMs), trained autoregressively on vast amounts of natural language data [19], exhibit emergent phenomena that endow the models with general intelligence [20], enabling them to simulate human behavior. Concurrently, as technological advancements progress, LLMs have evolved beyond generating high-quality natural language text, now demonstrating expertise comparable to that of human specialists in fields such as medicine [21], finance [22], marketing [23], and education [24].

To further enhance the effectiveness of LLMs as human assistants, a range of techniques have been developed to improve their reasoning ability, reliability, and interpretability. RAG integrates LLMs with external knowledge bases [25], thereby improving the quality of responses in specialized domains without requiring additional training. It also allows for dynamic updates, contributing to improved reliability. The CoT [26] reasoning technique enhances model transparency by requiring the system to output a sequence of intermediate reasoning steps before reaching a final answer. This enables users to observe the model's cognitive process directly, rather than merely its final conclusion, thereby providing valuable insights for human-assisted decision-making [27]. These measures significantly enhance reliability and reduce the occurrence of "hallucinations" (which represents incorrect or misleading outputs) [28].

C. Applications of LLMs in Medicine and Parkinson's Disease

LLMs have exhibited significant potential in the medical field [29]. Broadly, the applications of LLMs can be categorized into two primary categories: the first involves pretraining on extensive medical knowledge bases or performing In-Context Learning (ICL) within external medical knowledge environments. This approach enables the development of models capable of performing general tasks, such as medical record analysis, medical text summarization, and personalized healthcare assistance [30]. The second category focuses on utilizing individual patients' numerical data, such as continuous monitoring data from wearable devices, diagnostic and prescription records from electronic health records, to conduct health status classification and disease prediction [31]. These studies illustrate that by integrating the language comprehension capabilities of LLMs with expert knowledge from the medical domain, it is possible to enhance medical efficiency and diagnostic support to varying extents. In the domain of neurodegenerative diseases, particularly Parkinson's Disease (PD), researchers have begun investigating the potential of incorporating LLMs into disease management. For instance, Bouchouras et al. [32] developed a LLM-based model to assist with PD diagnosis and alert systems; Koga et al. [33] compared the diagnostic accuracy of ChatGPT and Google Bard in neurodegenerative pathology, though neither study can achieve treatment regimen generation. Cardenas et al. [34] enhanced detection systems by integrating LLM chatbots, which provided interactive preventive and rehabilitation advice for patients, though they did not offer personalized medication dosage recommendations. Additional efforts have focused on automating the analysis of large-scale data, such as the work by Stroganov et al. [35], who used LLMs to extract and standardize pathological reports from PD patients, significantly improving the efficiency of data analysis.

Overall, these efforts have established foundations for the application of LLMs in PD management. However, key aspects—such as optimizing medication dosage and frequency, personalized treatments, model interpretability and reliability, and updating medical knowledge—remain insufficiently explored. Therefore, developing LLMs for interpretable, clinically relevant, and medication-specific decision support systems in complex diseases is a pressing research priority.

III. PRELIMINARY

A. PD Treatment

The current standard treatment for Parkinson's Disease is Levodopa (L-dopa), which facilitates dopamine replacement and is generally regarded as the most effective therapy for PD. While Levodopa effectively reduces bradykinesia (slowness of movement), it can also induce dyskinesia (involuntary movements), particularly with long-term use. Levodopa-induced dyskinesia (LID) affects approximately 40–50% of patients within five years of treatment [36] and is believed to result from non-physiological, pulsatile dopaminergic stimulation [37]. In addition to impairing quality of life, LID complicates treatment planning due to its sensitivity to dose timing and absorption [38]. Therefore, effective Levodopa treatment requires a delicate balance of the contradictory effects of bradykinesia and dyskinesia.

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TABLE II
SUMMARY OF PPMI PATIENT CHARACTERISTICS

	All Patients	L-Dopa Type		
		IR	CR	Rytary
N	504	482	152	64
Female (%)	37.9	37.9	41.4	39.1
Age	65±10	65	64	64
Years PD	1.4±1.9	1.4	1.4	2.0
Initial MDS-UPDRS-III	23.6±12.0	23.7	23.0	22.4
MDS-UPDRS-III Change	4.9±13.8	4.9	4.7	3.9
Daily LED (mg)	528±328	536	633	738

Levodopa is available in several formulations: Immediate Release (IR), Controlled Release (CR), and Rytary (extended-release). IR is rapidly absorbed, providing quick symptom relief for acute exacerbations, but prolonged use may lead to dyskinesia. In contrast, CR is designed to prolong the release of the medication, resulting in a relatively slower onset of action, which may not be sufficient to promptly address acute flare-ups. Rytary combines both release mechanisms, providing fast relief initially, followed by sustained release to maintain the therapeutic effect. However, due to the complexity of Levodopa's release mechanism in Rytary, patients often experience fluctuations in drug efficacy, with periods of symptom relief followed by exacerbations [39]. As a result, optimizing Levodopa therapy for PD patients requires careful monitoring and drug adjustments. To address this challenge, there is a growing need for a data-driven framework to optimize medication management that can dynamically adjust treatment plans based on symptom tracking and individual patient profiles [40].

B. PPMI Dataset

The Parkinson's Progression Markers Initiative (PPMI) [41] is a large-scale observational study on PD, conducting long-term, periodic assessments of motor function and symptoms in over 700 patients. Follow-up visits were scheduled every three or six months, during which motor assessments were conducted. The distribution of follow-up intervals is detailed in Section A of the supplementary material (henceforth referred to as Supp. Material A). All patients received standard levodopa-based treatment. For our analysis, we selected 504 patients who had used specific levodopa formulations (L-Dopa IR, CR, or Rytary), excluding those who never used these formulations or had taken COMT inhibitors. Patients using additional non-levodopa medications (e.g., dopamine agonists, MAO-B inhibitors) alongside levodopa were retained, with non-levodopa dosages converted to levodopa-equivalent doses (LED) in mg using standard guidelines. These non-levodopa therapies were recorded in the covariates. Table II summarizes the descriptive statistics of the final cohort.

The dataset consists of three main components: Patient Covariates, Symptom Data, and Medication Data. **Patient Covariates** include information such as gender, age, non-levodopa dosages, and years since PD diagnosis. **Symptom Data** utilize the Movement Disorder Society–sponsored revision of the Unified Parkinson's Disease Rating Scale Part III (MDS-UPDRS-III) scores to assess motor impairment severity, ensuring consistency and comparability with Baucum et al. [17]. We excluded assessments marked as OFF-medication state or screening visits, but unrecorded medication states (NaN) in many records prevent confirming all remaining data as ON-medication. **Medication Data** records each patient's PD medication regimen, including type (IR, CR, Rytary), daily dosage, and administration frequency.

We cross-referenced the patients' covariates, symptoms, and medication records to generate time-series data corresponding to each patient's medication history.

C. Symptom Progression Prediction Model

Although Parkinson's disease treatment is a dynamic process without a definitive endpoint, we model it as a finite MDP with a discretized action space (limited medication regimens) and a finite number of medication cycles, enabling evaluation of treatment effects over a fixed time horizon. We model the patient's symptom progression as an MDP, defined as a tuple $\langle S, A, T\rangle$, where:

- S is the set of patient states, with $s_t \in S$ denoting the patient's symptom state at time t.
- A is the set of available treatment actions, with $a_t \in A$ denoting a certain medication combination taken at time t.
- T is a deterministic model that describes the patient's state transition. T maps the current state and action (s_t, a_t) to the next state s_{t+1}, with the patient's MDS-UPDRS-III score changing accordingly.

For the symptom progression prediction model, we adopt the **MDS-UPDRS-III prediction model** proposed by Baucum et al. [17], which employs an LSTM architecture. Specifically, this model utilizes pharmacokinetic modeling to convert daily medication dosage and frequency into a 35-dimensional vector [42]. This vector quantifies the blood drug concentrations measured at 30-minute intervals throughout 17 waking hours. Combined with regularized patient covariates and past scores, the model predicts the patient's MDS-UPDRS-III score for the next visit. Since both the input and output in the model require normalization, we scale the patients' scores to fit within a normalized range of 0 to 53, reflecting that 53 corresponds to the 99th percentile of the actual score distribution.

This progression prediction model can be mathematically expressed as follows:

$$x_t = x_{t-1} + \hat{\Delta x_t} + \epsilon,$$
$$\hat{\Delta x_t} = f\left([\Delta x_{(t-t_{\text{past}}):(t-1)}, \ell_{t-1}, \mathbf{c}]^T\right),$$

where:

- x_t is the (arctanh-transformed) MDS-UPDRS-III score at time t.
- $\Delta x_t = x_t - x_{t-1}$ denotes the change in the score, i.e., the difference between the scores at time t and $t-1$.
- $\hat{\Delta x_t}$ represents the predicted change in the score.
- ϵ is a Gaussian disturbance term that follows a normal distribution with a mean of 0 and standard deviation σ.

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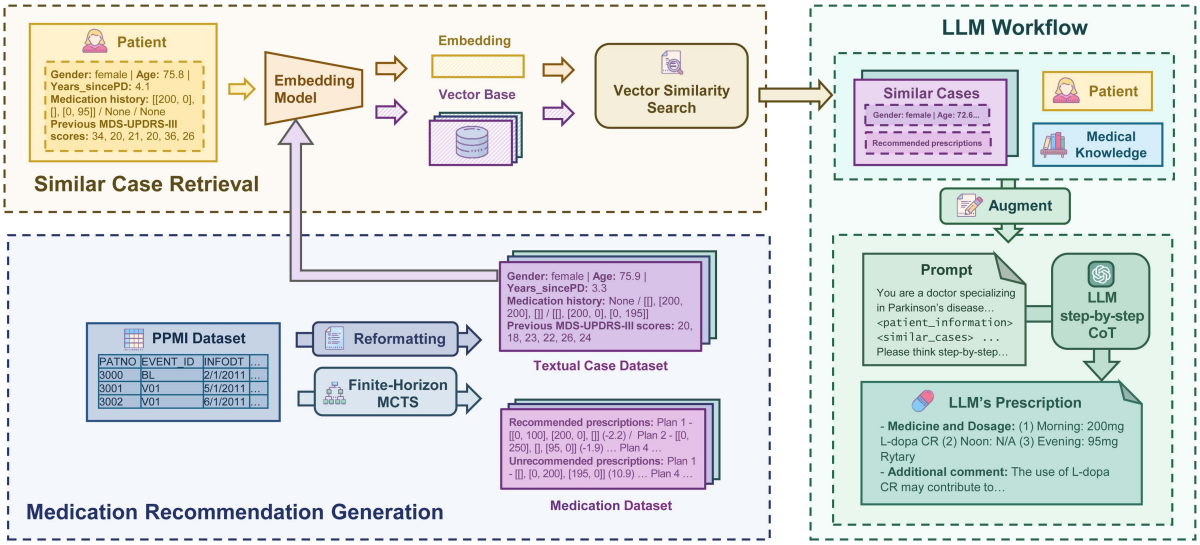

Fig. 1. The overall architecture of the PD-LLM framework. The **Medical Recommendation Generation** process employs Finite-Horizon MCTS to generate medication recommendations based on a finite patient dataset. **Similar Case Retrieval** computes patient similarity scores using embeddings from input profiles and historical case database. Next, the RAG process identifies relevant cases, followed by **LLM workflow** to generate personalized treatment recommendations based on medical guidelines and clinical evidence.

- ℓ_{t-1} is the L-dopa profile of the patient at time $t-1$, which quantifies the concentration levels of Levodopa in the blood.
- $\mathbf{c}$ is a vector of covariates, including gender, age, years since PD diagnosis, and dosages of the non-levodopa medications expressed as the levodopa-equivalent doses (LED) in milligrams (mg).

IV. Methodology

In this section, we propose a framework named **PD-LLM**, designed to use LLMs to provide personalized medication recommendations for PD patients. As illustrated in Fig. 1, the PD-LLM framework consists of three key processes:

- *Medication Recommendation Generation:* Generation of the probabilistically optimal medication recommendation dataset utilizing Finite-Horizon MCTS (Subsection A).
- *Similar Case Retrieval:* Knowledge retrieval based on vector similarity for retrieving clinically relevant cases (Subsection B).
- *LLM workflow and CoT Reasoning:* Step-by-step reasoning to improve the interpretability of medication strategies (Subsection C).

The workflow begins with textually processing original patient cases, converting clinical data and medication records into formats suitable for LLM processing. Using the Finite-Horizon MCTS method, we employ the symptom prediction model to simulate disease progression and identify optimal medication strategies. These strategies are compiled into a medication recommendation dataset and incorporated into the vector knowledge base of the RAG system. When a patient seeks treatment, the system calculates vector similarity to retrieve relevant cases from the knowledge base, which include recommended treatment plans. The input for the LLM consists of three key components: textually processed patient records, similar cases, and authoritative Parkinson's disease guidelines. This retrieved information is integrated into the prompt alongside CoT reasoning, enabling the LLM to perform step-by-step analysis and generate comprehensive medication recommendations accompanied by detailed explanations.

A. Finite-Horizon MCTS

Before integrating LLMs into medication decision-making, we employ machine learning-based prediction methods to support and refine the process. Although LLMs utilize extensive prior knowledge to provide empirically accurate recommendations, relying solely on experience-based suggestions may lead to errors. To address this, we incorporate MCTS [43], a probabilistic method for optimizing solutions in finite decision spaces with predefined endpoints through limited simulations.

Specifically, we construct a tree structure in which the root node represents the patient's current state. During each simulation, a medication combination is selected from the action space. The prediction model is then used to predict the patient's MDS-UPDRS-III score at the next follow-up visit under the selected plan. The simulation transitions to a new state, represented as children node of the current root, and the process continues recursively to simulate three consecutive medication administrations.

Based on the patient's current state information, we conduct randomized simulations of medication strategies. To evaluate both the short-term and long-term effects, we use the expected MDS-UPDRS-III score change after three follow-up visits as the evaluation criterion. After iterative rollout simulations, the

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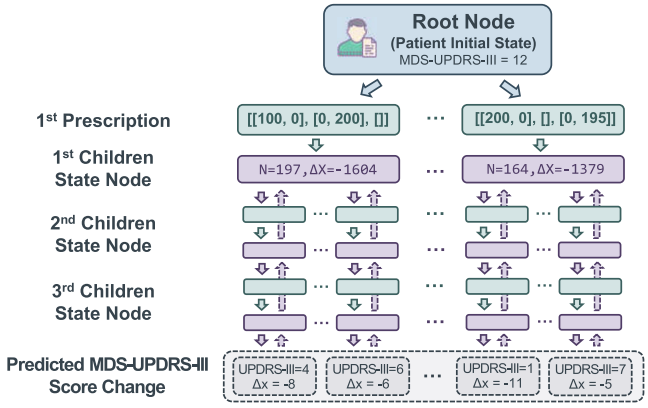

Fig. 2. A simple illustration of Finite-Horizon MCTS. MCTS selects actions based on the UCT priority score, performing iterative rollouts and simulations to estimate expected MDS-UPDRS-III scores for PD medication prescriptions.

patient's score change, relative to their initial state, is calculated and back-propagated through the tree structure. Fig. 2 illustrates this process, with pseudocode presented in Algorithm 1. Notice that in Algorithm 1, each medication action is linked to its resulting unique patient state due to the deterministic assumption. Although MCTS uses deterministic transitions, its repeated sampling across multiple time steps introduces variability through diverse branches, capturing uncertainty in symptom progression. Thus, we extend the traditional Upper Confidence Threshold (UCT)-based MCTS into a finite-horizon version tailored for PD treatment. Specifically, the priority score of each action during single-step plan selection is determined by the following formula:

$$\text{UCT}|_{s' \to T(\mathbf{LSTM}, s, a)} = -\frac{\Delta \text{X}(s')}{N(s')} + C_p \sqrt{\frac{2 \ln N(s)}{N(s')}},$$

where:

- $N(s)$ and $N(s')$ are counters that record the number of times the current state node and its children state node have been reached in previous simulations, respectively.
- $\frac{\Delta \text{X}(s')}{N(s')}$ is the expected MDS-UPDRS-III score change at the children state node.
- C_p is a hyperparameter balancing exploration and exploitation.

To ensure sufficient exploration of all nodes, the UCT value is set to positive infinity for actions leading to child nodes with $N(s') < 10$. This mechanism prioritizes under-visited child nodes for random selection and further exploration, ensuring sufficient exploration across the action space.

Nevertheless, the MCTS approach has several limitations. Specifically, MCTS requires extensive simulations for each single-step decision, repeatedly invoking the prediction model to estimate the expected MDS-UPDRS-III scores. This not only incurs significant computational costs but also makes the algorithm highly dependent on the accuracy of the prediction model, and may introduce cumulative errors. Moreover, MCTS lacks generalizability, as it cannot share treatment experiences across different patients. Its design is primarily suited to discrete action spaces, making it challenging to handle continuous action spaces, which limits its applicability in designing flexible medication regimens.

To mitigate these issues, we propose a workflow that combines MCTS with LLM. First, MCTS operates within a discretized action space via UCTguided simulations, with the search depth and number of rollouts bounded to ensure computational feasibility. Child nodes with higher UCT values (typically indicating a larger expected decrease in MDS-UPDRS-III score) are allocated more visits, while less promising nodes receive fewer, thereby improving search efficiency. After the offline simulations, the top treatment trajectories are assembled into a medication recommendation dataset and stored in a vector knowledge base. Subsequently, the LLM combines this dataset with patientspecific features to convert MCTS's discrete suggestions into finegrained, continuous prescriptions, automatically tuning dosage and frequency for personalized PD treatment.

Algorithm 1: Finite-Horizon MCTS for PD Prescription Searching.

Require: Root node s_0, action space A, MDS-UPDRS-III Prediction Model **LSTM**

function BackPropagateNode s, score change Δx
  while s is not None **do**
    $N(s) \leftarrow N(s) + 1$
    $\Delta X(s) \leftarrow \Delta X(s) + \Delta x$
    $s \leftarrow$ parent of s
  end while
end Function

function GetBestChildNode s
  return $\text{argmax}_{s' \leftarrow T(\mathbf{LSTM}, s, a), a \in A}$ $-\Delta X(s')/N(s') + C_p * \sqrt{2 * \ln N(s)/(N(s') * \text{bool}(N(s') > 10))}$
end Function

function ExpandNode
  for $t = 0 \to t = 2$ **do**
    $s_{t+1} \leftarrow$ GetBestChild(s_t)
  end for
end Function

for Number of simulations **do** ▷ Main Process
  $s_3 \leftarrow$ ExpandNode(s_0)
  Obtain the MDS-UPDRS-III score difference Δx between s_3 and s_0
  BackPropagate(s_3)
end for
Evaluate an action based on the value of $\Delta X/N(s_1)$ of its corresponding transition state, for any $s_1 \in$ children of s_0

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B. Vector Knowledge Base and Similarity Search

1) Structuring Patient Information and the Medication Dataset: We use concise and highly readable natural language to describe patient information, including key elements such as age, gender, clinical history, and medication history. Each case is simulated using Finite-Horizon MCTS. Based on the expected MDS-UPDRS-III score change, the search space is filtered to extract the four most recommended and four least recommended medication regimens, which form the medication recommendation dataset. Each regimen is represented in a structured Python list format, accompanied by its corresponding expected MDS-UPDRS-III score. This structured format enables LLMs to analyze medication regimens, accounting for factors such as dosage and frequency. By offering multiple alternative regimens, this approach enhances the model's ability to evaluate options and generates more precise, personalized, and clinically informed treatment recommendations.

2) Similar Case Retrieval Based on Text Embedding Similarity: To complete the RAG workflow, we identify the most relevant similar cases and their corresponding treatment recommendations by evaluating the similarity between a given patient and other cases in the dataset. The detailed steps are as follows:

Text Embedding Generation: A pre-trained text embedding model is used to encode the target patient's information into an embedding vector, denoted as $\mathbf{emb}_p$. Similarly, the information of all cases in the dataset is encoded to construct a vector knowledge base, represented as $\mathbf{emb}_b$.

Similarity Calculation: Cosine similarity is widely used in RAG frameworks to rank documents based on embedding similarity scores [44]. Its insensitivity to vector magnitude and focus on directional similarity make it well-suited for semantic matching in RAG systems [45], [46]. In our approach, cosine similarity measures the similarity between the target patient's embedding vector $\mathbf{emb}_p$ and the embedding vectors of all cases $\mathbf{emb}_b$, calculated as follows:

$$\text{similarity} = \frac{\mathbf{emb}_p \cdot \mathbf{emb}_b}{\|\mathbf{emb}_p\| \|\mathbf{emb}_b\|}.$$

The resulting similarity score ranges between 0 and 1, where higher scores indicate greater similarity.

Selection of Similar Cases: To ensure accurate matching, we incorporate not only structured patient attributes (e.g., age, gender), but also medication history and previous MDS-UPDRS-III scores as part of the input to the embedding model. From the dataset, the two cases with the highest similarity scores are identified and their corresponding treatment information is retrieved. These selected cases are then incorporated into the input prompt for the LLM, serving as references for making medication decisions. Technical details can be found in Supp. Material F.

By adopting this approach, we integrate the target patient's individual characteristics with similar cases. This method leverages vector similarity retrieval to provide high-quality reference information for the LLM, thereby significantly improving the precision and personalization of its medication recommendations.

C. Prompt Design

The input prompt integrates key components as follows: textually processed patient records, similar cases retrieved from a vector knowledge base, and evidence-based treatment guidelines, as illustrated in Fig. 3.

The knowledge base, detailed in Section IV.B, contains patient embeddings and MCTS-generated medication recommendations, enabling RAG to identify relevant cases via vector similarity. Treatment guidelines, such as the AAFP Parkinson Disease guideline [47], which specifies recommended medication types and dosage ranges, provide authoritative medical knowledge. These guidelines are incorporated into the prompt to ensure clinical relevance and alignment with established standards. For efficiency, concise guidelines are currently included directly in the prompt. This architecture maintains scalability, permitting future expansion to larger datasets that include resources such as medical literature, with targeted integration achieved through adaptive retrieval mechanisms.

To address task-specific requirements, we developed a step-by-step, domain-knowledge-driven prompt leverages the LLM's semantic understanding to synthesize patient data, similar cases, and guidelines, following a clear reasoning process to generate outputs [48].

The prompt design includes the following components:

Structured Descriptions for Patient and Similar Case Information: The prompt includes concise and structured natural language descriptions of the target patient and similar cases, along with explanations of their potential influence on treatment, which help the LLM understand the contextual background and apply relevant medical knowledge effectively.

Step-by-Step Chain-of-Thought Reasoning: The LLM analyzes patient information, examines similar cases and their suggested regimens, and verifies their relevance. Subsequently, it balances multiple factors to provide a comprehensive explanation and generate tailored medication recommendations, guided with medical knowledge context.

Structured Output for Medication Plans: The medication plans generated by the LLM are presented in a textual format, which includes frequency, dosage, as well as a clear explanation of the rationale behind the plan's design. The output is then converted into a structured Python list format to facilitate automated processing and ensure clarity.

Overall, this framework integrates patient-specific and similar-case data with explicit reasoning and professional expertise. By guiding the LLM through a systematic reasoning process and stepwise outputs, it ensures a complete reasoning chain and provides personalized, reliable, and interpretable medication recommendations.

V. Experiment

A. Data Processing

The original PPMI dataset contains 504 patients who used specific formulations of Levodopa, with each patient undergoing k follow-up visits. Due to variations in the number of follow-ups across patients, we retained only those patients with $k \geq 7$. This

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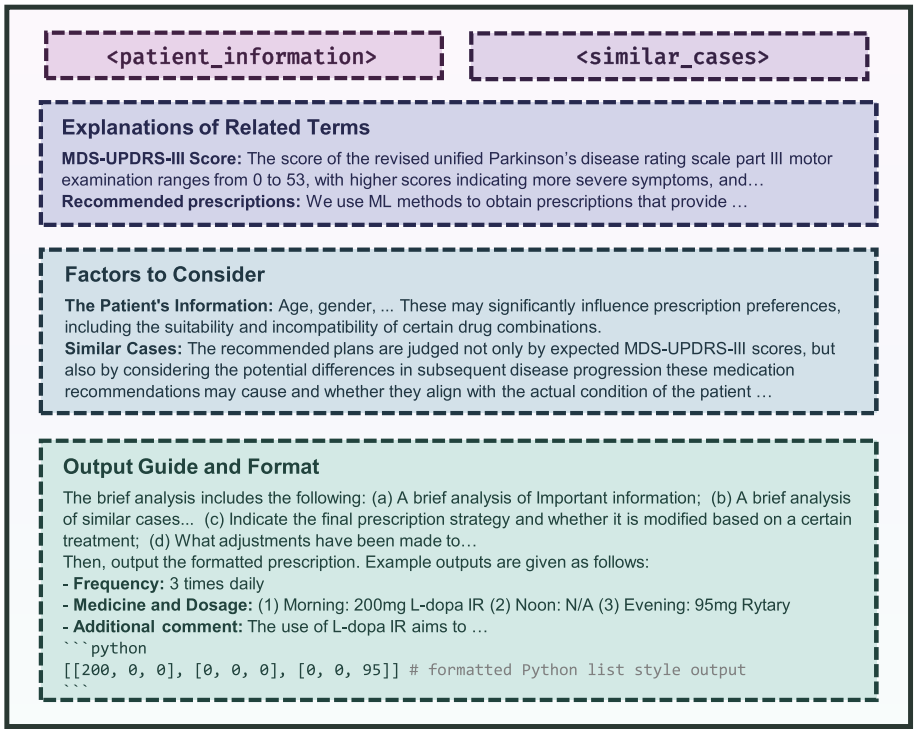

Fig. 3. Simplified prompt example for the PD-LLM framework. The prompt integrates processed patient records, similar cases, and evidence-based treatment guidelines to provide clinical context for LLM decision-making. It also includes output format instructions to guide the LLM in generating step-by-step reasoning and personalized medication prescriptions.

filtering resulted in a final cohort of 399 patients, representing approximately 79% of the original dataset. The average interval between treatments for these patients ranged from 4 to 12 months. To make the most of limited data while preventing data leakage, we employed Leave-One-Out Cross-Validation (LOOCV), a specific form of k-fold cross-validation where k equals the total number of patients in the dataset. During the testing phase, the PD-LLM framework generates medication strategies for each individual patient, utilizing a similar-case database that exclusively contains historical data from the remaining 398 patients.

B. Implementation Details

In implementing the PD-LLM workflow, we focused on the following key aspects:

LLM Selection and Parameter Configuration: We selected the DeepSeek-v2.5 model [49] as the base LLM, with parameters set to `temperature = 0.2` and `Top P = 1` to ensure deterministic and consistent text generation. This selection was based on DeepSeek-v2.5's exceptional performance on widely recognized benchmarks for large language models and its cost-efficiency with low cost per million tokens. To strengthen the empirical basis for our model choice, we conducted comparative evaluations with various LLMs such as DeepSeek-v3 and Qwen, revealing marginal performance differences among general-purpose LLMs (see Supp. Material D for details).

MCTS Configuration: The MCTS action space comprises 73 medication regimens, each defined by a specific combination of drug types, dosages, and dosing frequencies (provided in Supp. Material B). The dataset included 399 patients, contributing a total of 2,000 unique states, as patient states varied across follow-up visits. For each patient state, MCTS simulations were employed to predict the expected change in MDS-UPDRS-III scores after 3 follow-up visits, optimizing treatment strategies to construct the medication recommendations dataset. Each state underwent 2,500 simulations, with the number of simulations determined through a convergence analysis of the Monte Carlo method to ensure stability and reliability. The MCTS logic, hyperparameter settings, and robustness analysis of the simulation process are provided in Supp. Material C.

Similar Case Matching: We employed the MedEmbed-large embedding model [50] for similar case matching to ensure relevance and accuracy in the retrieved results. MedEmbed-large has demonstrated superior performance in medical text embedding tasks, effectively capturing the semantic similarity between cases. During the matching process, records from the same patient (identified by the same PATNO) were excluded to prevent data leakage and overfitting.

Prediction and Evaluation: We used the prediction model to predict the expected change in MDS-UPDRS-III scores and evaluate the long-term therapeutic outcomes of different medication regimens, which has been rigorously validated and is recognized for its high predictive accuracy.

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C. Evaluation Metrics

This study evaluates treatment effectiveness using two distinct sets of metrics: overall treatment outcomes and comparative improvements over physician-administered treatments.

For overall treatment outcomes, we use the **MDS-UPDRS-III Change** metric to evaluate effectiveness for the overall test set, which is defined as the average change in MDS-UPDRS-III scores across all patients and is represented as:

$$\Delta U = \frac{1}{N}\sum_{i=1}^{N}\left(U_{k_i}^{(i)} - U_0^{(i)}\right),$$

where N is the total number of patients included in the study, $U_0^{(i)}$ is the baseline MDS-UPDRS-III score at time point $t = 0$ of patient i, and $U_{k_i}^{(i)}$ is the score of patient i at their last follow-up visit k_i.

For each patient, a new medication strategy is implemented after each follow-up visit, and the prediction model predicts the MDS-UPDRS-III score at the next visit. This process continues until k follow-up visits to calculate the total score difference through the whole process. This allows for a direct comparison of the score change (ΔU) across the proposed medication strategies. As a "lower is better" metric, a decrease in ΔU indicates reduced symptom severity, given that higher scores reflect worse motor impairments. We report the mean, median, 25th percentile, and 75th percentile of ΔU across the test set for each method. Additionally, we conduct Paired-Sample t-tests to assess whether the observed treatment improvements are statistically significant.

To further evaluate the effectiveness of each method relative to physician-administered treatments, we introduce two additional patient-level metrics: (1) the proportion of patients showing any improvement in MDS-UPDRS-III scores compared to physician recommendations (denoted as **with MDS-UPDRS-III reduction**), and (2) the proportion of patients achieving a reduction of at least 2 points (denoted as **with >2-point reduction**), which serves as a threshold for individual-level benefit suggested by prior work [17]. Specifically, for each patient, we compare the ΔU achieved under the proposed method versus that of the physician-prescribed regimen. We then compute the percentage of patients who meet either criterion across the test set.

D. Baselines

To evaluate the effectiveness of our proposed framework, we designed the following three baseline approaches. Specifically, comparisons between the LLM-method and the Physician-method demonstrate the effectiveness and potential value of LLMs in designing treatment strategies. Additionally, comparisons between the LLM-method and the RL-method highlight the advantage of leveraging both numerical and textual information.

Real: The actual MDS-UPDRS-III score change are recorded in the PPMI dataset following physician-administered treatment.

TABLE III
KEY METRICS FOR MDS-UPDRS-III CHANGE (ΔU) UNDER DIFFERENT EXPERIMENTAL SETTINGS

Metrics	Average↓	Median↓	25th percentile↓	75th percentile↓
Real	5.17	4.00	-4.00	14.00
Physician	3.26	3.46	-2.98	10.06
RL	2.84	3.15	-2.05	8.72
PD-LLM	**1.83**	**1.86**	**-3.83**	**8.14**

Physician-method: This method uses the prescriptions provided by physicians in the PPMI dataset, predicting MDS-UPDRS-III changes utilizing the prediction model. It represents current clinical practices and serves as a highly relevant benchmark for real-world reference. While discrepancies between the Physician-method and real outcomes are inevitable due to inherent prediction errors in the MDS-UPDRS-III prediction model, the score change predicted by the Physician-method remain comparable to those of the RL and PD-LLM methods.

RL-method: This baseline employs the RL model proposed by Baucum et al. [17], which generates medication recommendations by leveraging an action space of potential treatment combinations. A ten-fold cross-validation process is applied to the dataset of 399 patients to prevent data leakage. The model is rigorously trained and validated to explore long-term treatment strategies. Details of the RL model's configuration are provided in Supp. Material B.

These baseline methods represent both current clinical practices and advanced data-driven AI approaches, enabling an effective comparison to demonstrate the superiority of our proposed PD-LLM framework in optimizing personalized long-term treatment outcomes for PD patients.

E. Main Results

Table III presents four metrics of MDS-UPDRS-III Change (ΔU) under different experimental settings. The results indicate that the PD-LLM approach achieves an average improvement of 1.43 points in ΔU compared to the Physician group, whereas the RL method shows a smaller improvement of 0.42 points relative to the Physician group. Furthermore, PD-LLM outperforms across all statistical metrics, indicating significant overall improvements. Besides, a paired-sample t-test analysis confirms that the improvement by the RL-method over the Physician-method is statistically significant ($p = 0.029$), while the improvement achieved by the PD-LLM is highly statistically significant ($p = 3.2 \times 10^{-9}$).

F. Ablation Study

To validate the effectiveness of the PD-LLM workflow and assess the contribution of each component, we designed four ablation experiments. The primary components of PD-LLM include RAG and CoT. Within RAG, two critical subcomponents are MCTS and Vector Similarity Search (VSS).

The four ablation experiments are designed as follows:

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TABLE IV
ABLATION STUDY: KEY METRICS OF MDS-UPDRS-III CHANGE (ΔU) FOR PD-LLM WORKFLOW COMPONENTS

Metric	Average↓	Median↓	25th percentile↓	75th percentile↓
Physician	3.26	3.46	-2.98	10.06
w/o RAG	2.68	3.09	-3.52	9.04
w/o MCTS	2.71	2.75	-2.99	8.50
w/o VSS	2.40	3.00	-3.50	8.71
w/o CoT	2.26	2.80	-3.42	9.03
PD-LLM	**1.83**	**1.86**	**-3.83**	**8.14**

TABLE V
METRICS OF TREATMENT EFFICACY IMPROVEMENTS FOR RL AND VARIOUS LLM WORKFLOWS VS. PHYSICIAN SETTING

Metrics	with MDS-UPDRS-III reduction (%) ↑	with >2-point reduction (%) ↑
RL	52.6	24.3
w/o RAG	53.4	34.8
w/o MCTS	54.6	34.3
w/o VSS	54.4	35.3
w/o CoT	60.1	39.1
PD-LLM	**61.7**	**43.4**

- *Without RAG (**w/o RAG**):* This configuration removes all RAG components, requiring the LLM to directly analyze patient information using CoT reasoning to provide medication strategies and explanations. This experiment evaluates the overall impact of RAG on performance.
- *Without MCTS (**w/o MCTS**):* This configuration retains VSS but replaces the MCTS medication plans with recommendations generated by the RL policy network, aiming to assess the contribution of MCTS.
- *Without VSS (**w/o VSS**):* This configuration retains MCTS but replaces VSS with random sampling of reference cases from the medication recommendation dataset. This evaluates the contribution of VSS.
- *Without CoT (**w/o CoT**):* This configuration retains all RAG components but removes the CoT prompts. The LLM provides only medication dosages without reasoning or explanations. This experiment assesses the impact of CoT on the overall workflow.

To ensure the robustness of the ablation study, we performed 100 repetitions of the k follow-up visits for each patient. This thorough evaluation confirms the importance of each component in PD-LLM and establishes a clear hierarchy of their contributions. These results provide valuable insights for future optimization and extension of PD-LLM, ensuring both the reliability and consistency of its performance.

Table IV presents the variations in MDS-UPDRS-III Change across different ablation groups, emphasizing the importance of each component in the PD-LLM framework. IQRs are reported for their robustness, and corresponding standard deviations are provided in Supp. Material H.3. RAG emerges as the most critical component of PD-LLM, playing a pivotal role in integrating external knowledge and optimizing medication strategies. Its removal results in the most significant performance decline. Among the RAG subcomponents, MCTS and VSS contribute significantly to improving treatment outcomes by optimizing medication plans and ensuring the relevance of reference cases, respectively. Our cosine similarity–based retrieval method, implemented in VSS of full "PD-LLM", significantly outperforms "w/o VSS" setting, demonstrating this specific retrieval strategy is effective and high-quality. Even under "w/o VSS" (random retrieval), the LLM outperforms the physician baseline and the "w/o RAG" setting (no retrieval case), indicating robustness to retrieval noise and reliable performance with lower-quality inputs. Although CoT has a relatively smaller impact on score improvement, its ability to explain medication recommendations enhances user understanding and trust in the AI system, thereby indirectly contributing to better treatment results.

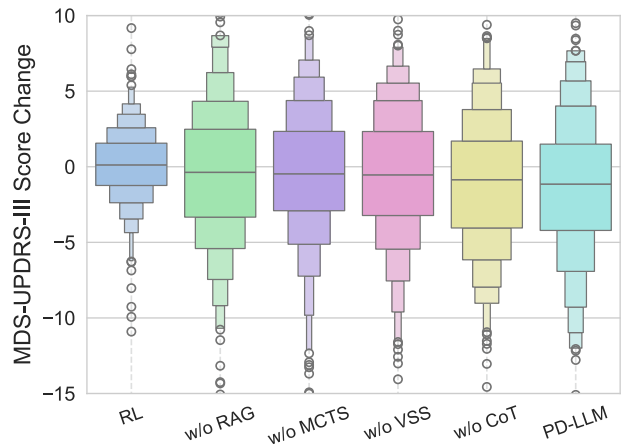

Fig. 4. Enhanced box plot of MDS-UPDRS-III score change for RL and various LLM workflows vs. Physician setting.

Additionally, Table V shows the metrics of treatment efficacy improvements when compared to the Physician setting, and Fig. 4 shows the corresponding enhanced box plot of MDS-UPDRS-III score change. Table V shows that PD-LLM can achieve better therapeutic outcome than physicians in 61.7% of patients, and at least 2-point decrease in MDS-UPDRS-III scores in 43.4% of patients, remarkably outperforming the RL setting mentioned above.

To further support the reliability and robustness of the proposed workflow, we conducted several complementary analyses. Specifically, we evaluated the VSS module across a heterogeneous patient cohort with diverse progression trajectories, confirming stable performance and its key contribution to PD-LLM (see Supp. Material H.1). We also analyzed retrieval robustness in edge cases and outliers to ensure safe performance under high-risk conditions (see Supp. Material H.2). Finally, we conducted subgroup performance across Physician, RL, and PD-LLM settings to support equity claims (see Supp. Material E).

VI. DISCUSSION

A. Evaluation and Comparison of Treatment Regimens

To evaluate medication preference differences among the PD-LLM, Physician, and RL methods, we conducted a comparative

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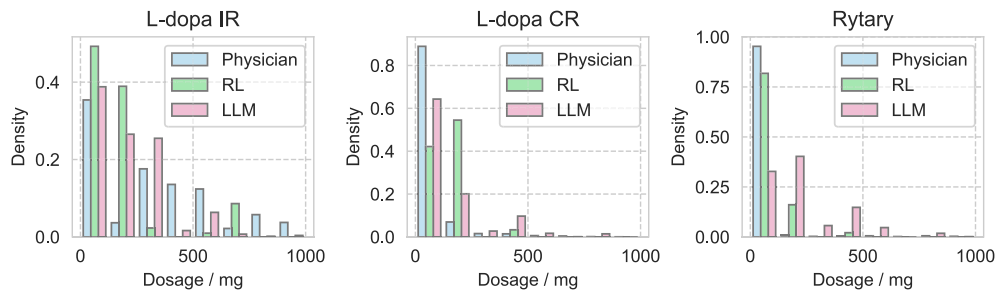

Fig. 5. Distribution of daily Levodopa dosages under different treatment methods.

TABLE VI
SUMMARY OF MEDICATION USAGE UNDER DIFFERENT EXPERIMENTAL SETTINGS

	Physician	RL	LLM
Taking IR (%)	66.7	78.8	72.9
Taking CR (%)	14.7	72.5	51.1
Taking Rytary (%)	4.7	31.2	86.3
IR daily dose (mg)	556	224	244
CR daily dose (mg)	216	189	239
Rytary daily dose (mg)	948	167	246
Doses per day	3.29	3.03	3.05

analysis focusing on medication proportions and dosages, as presented in Table VI, and Fig. 5 illustrates the dosage distribution for each method.

1) Medication Proportions and Dosage: The Physician group predominantly relied on L-dopa IR (66.7%), with much lower usage of L-dopa CR (14.7%) and minimal use of Rytary (4.7%). Furthermore, in approximately 20% of follow-up records, physicians opted not to prescribe any medication, reflecting a more conservative treatment approach. And they typically favored single-drug regimens. In contrast, both the RL and PD-LLM methods demonstrated greater ability to implement combination therapies that leverage multiple medications to optimize treatment outcomes. Notably, the PD-LLM group increased the usage of Rytary (86.3%), far surpassing the Physician group (4.7%) and the RL group (31.2%).

Regarding medication dosages, the Physician group administered the highest average daily dosages across patients. In contrast, due to its limited discrete action space, the RL group's dosage distributions appeared as isolated spikes on the histogram, a consequence of the few action options available. It primarily used a combination of L-dopa IR and CR, occasionally supplemented with high-dose IR or low-dose Rytary. By comparison, the PD-LLM offers greater flexibility in dosage adjustments, dynamically tailoring medication proportions to optimize outcomes based on patient needs, resulting in a more continuous histogram distribution. It allows for more precise dose adjustments, enhancing the personalization and accuracy of treatment strategies.

2) Discussion and Insights: The PD-LLM framework notably increased the use of Rytary, whose extended-release formulation provides a smoother pharmacokinetic profile. This enhancement minimizes fluctuations in drug efficacy, ensuring more stable therapeutic effects. However, its use requires meticulous dose adjustments and careful monitoring. Also, Rytary's limited adoption in physician-prescribed regimens may stem from its recent FDA approval in 2015.

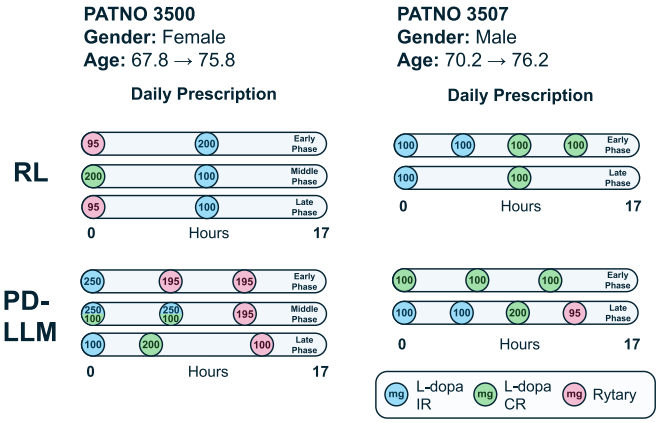

Fig. 6. Basic information and daily prescriptions given by RL and PD-LLM for two representative patients, including medication type, dosage, and schedule over a 17-hour day (from 5:00 am to 10:00 pm).

In addition, PD-LLM demonstrated remarkable flexibility in personalized combination therapies by using L-dopa IR or CR during the daytime and supplementing with Rytary at night. This tailored approach not only optimizes the overall therapeutic efficacy but also improves patients' quality of life, as further detailed in the case study in Section V. Overall, PD-LLM exhibited distinct advantages in facilitating combination therapies, enabling flexible dosage adjustments, and designing personalized treatment regimens, thereby offering promising advancements for Parkinson's Disease management.

B. Case Study

To provide a more detailed and intuitive understanding of the medication strategies devised by PD-LLM and its differences compared to the Physician and RL groups, we conducted a case study on two representative patients from the PPMI dataset, identified as 3050 (PATNO 3050) and 3057 (PATNO 3057), and their key demographic and daily prescriptions are outlined in Fig. 6. The daily medication recommendations from RL and LLM contain the medication type, dose, and schedule at different disease progression phases.

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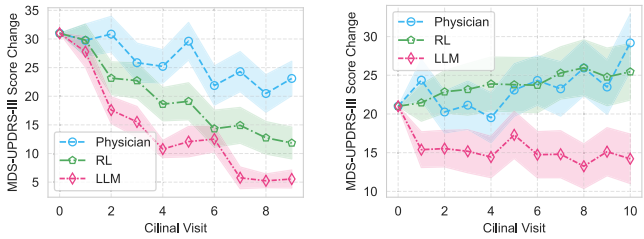

(a) Patient 3500 (b) Patient 3507

Fig. 7. Trends of MDS-UPDRS-III scores over time for two representative patients under different experimental settings.

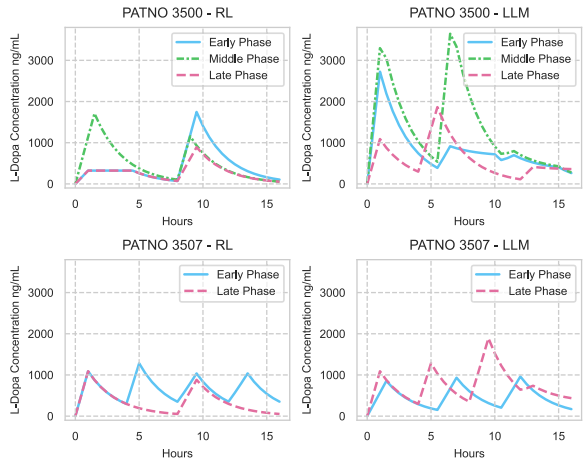

Fig. 8. Daily L-dopa profile at different clinic visit for two representative patients under different experimental settings.

Fig. 7 illustrates the predicted trends in MDS-UPDRS-III scores under different methods. As shown, Patient 3050 demonstrated stable disease control over time, with PD-LLM achieving better therapeutic outcomes compared to other methods. Conversely, Patient 3057, who exhibited a worsening disease trajectory, showed improvement with PD-LLM's more effective symptom progression mitigation. Also, Fig. 8 shows the corresponding daily L-dopa profiles for both patients, charting the blood concentrations of L-dopa throughout the day (from 5:00 am to 10:00 pm). It details the profiles for patient 3500 at the early (2nd clinical visit), middle (6th clinical visit), and late phase (9th clinical visit), as well as for patient 3507 at the early (1st clinical visit) and late phase (6th clinical visit).

For patient 3500, during the early and middle stages, due to the patient's poor condition, LLM increased L-dopa blood concentrations by administering higher doses of L-dopa IR and CR in the morning. In the late stage, as the patient's condition improved, the dosages were appropriately reduced. Throughout the treatment process, LLM consistently provided Rytary in the afternoon to maintain stable blood drug levels and ensure continuous therapeutic effects. For patient 3507, whose initial MDS-UPDRS-III score was low, LLM administered continuous low doses of L-dopa. In the late stage, with the onset of fluctuations and a tendency for deterioration, LLM increased the dosage of L-dopa CR and supplemented it with Rytary to stabilize the blood drug concentration.

These case studies highlight the adaptability and efficacy of PD-LLM in managing varying disease stages and patient-specific needs, further underscoring its potential to improve personalized Parkinson's Disease management.

C. Clinical Significance and Insights

Clinical Relevance: The proposed framework demonstrates significant clinical relevance, particularly through its focus on interpretability and similarity analysis in medical practice.

By incorporating CoT reasoning, our system enhances interpretability, enabling physicians to evaluate treatment validity, detect potential errors, and ensure transparent decision-making. First, since medical decisions directly impact patient safety, CoT-generated explanations enable physicians to quickly evaluate the validity of the model's decisions, avoiding reliance on closed box outputs. Additionally, CoT reasoning reveals potential flaws in the model's decision-making process, allowing researchers to trace and rectify errors. This step-by-step reasoning has also demonstrated performance improvements in models such as GPT-o1 and Deepseek-R1. Furthermore, in high-risk domains like healthcare, interpretability ensures compliance with ethical and legal standards by making the decision-making process transparent, thereby mitigating risks of bias, discrimination, or legal violations.

Additionally, similarity analysis enhances treatment quality and system efficiency. By retrieving cases similar to the current patient's profile from a large case database, the model leverages evidence-based treatment plans that have already proven effective. This reduces the risk of incorrect or misleading outputs from the AI system and increases the relevance and effectiveness of the proposed strategies, leading to improved patient outcomes and higher-quality recommendations. Moreover, similarity analysis minimizes training costs by relying on a continually expanding and updated clinical case database. Instead of requiring frequent retraining or adjustments to the model's core architecture, the system integrates the latest clinical data and research findings by simply updating the database. This dynamic integration ensures that the model remains current with evolving medical knowledge while maintaining operational efficiency.

Innovation: To the best of our knowledge, this is the first LLM-based system for personalized PD treatment. Our framework combines structured data and textual inputs to generate treatment plans. It outperforms traditional closed box RL approaches by integrating MCTS-guided dose optimization, RAG-based clinical retrieval, and CoT reasoning. Each component addresses a key clinical challenge: MCTS enables fine-grained exploration of medication dosages, RAG retrieves context-relevant evidence from evolving clinical knowledge, and CoT facilitates stepwise interpretability of treatment decisions. This design closely aligns with real-world PD treatment needs by enabling multi-drug regimens and tailored dose adjustments, and provides an effective, scalable, and transparent solution for digital PD care.

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D. Limitations

Limitations of the PPMI Dataset: While the PPMI dataset provides valuable insights for our study, its limitations impact the generalizability and clinical applicability of our PD-LLM framework. First, the dataset primarily includes patients using specific levodopa formulations (L-Dopa IR, CR, and Rytary). Our analysis focused solely on levodopa adjustments (formulation, dosage, and frequency), neglecting concurrent use or dose changes of other PD medications, which restricts applicability to broader treatment regimens.

Second, key clinical and socioeconomic factors are underrepresented in the data. For example, gastrointestinal (GI) dysmotility, a common PD comorbidity affecting levodopa efficacy, is inadequately captured. Similarly, socioeconomic factors, such as medication costs and insurance coverage, are incompletely documented and excluded from our model, potentially biasing recommendations across diverse populations.

Given these constraints, future work should incorporate more comprehensive datasets, including gastrointestinal metrics, multi-drug regimens, and cost data. These limitations highlight the need for cautious interpretation of our model's recommendations and further validation in varied clinical contexts.

Ethical Implications of LLM-Based Treatment: The application of LLMs in personalized medicine for Parkinson's disease raises several important ethical considerations. First, data privacy is a fundamental concern, as LLMs process sensitive patient information. Although our framework uses anonymized data, real-world implementation must comply with regulations like HIPAA and GDPR to prevent data breaches or misuse of personal health information.

Second, algorithmic bias poses a risk to the fairness of treatment recommendations. The training data used in our framework, namely the PPMI dataset, may not fully represent the diversity of real world patient populations, particularly in terms of ethnicity, socioeconomic backgrounds, and comorbid conditions. Consequently, the model's recommendations could potentially lead to biased outputs that disadvantage underrepresented groups. Future work should prioritize diverse datasets to enhance equity and generalizability.

Finally, human oversight is essential to avoid over-reliance on LLM-generated recommendations. While our PD-LLM framework provides tailored insights, it outputs are inherently probabilistic and may overlook subtle clinical cues or individualized factors that experienced clinicians can recognize. To enhance reliability, we envision integrating human-in-the-loop oversight with formal uncertainty quantification and explainability auditing. These mechanisms can highlight low-confidence predictions and flag potential inconsistencies, while preserving clinician authority in the decision-making process. This approach ensures patient safety, maintains accountability, and fosters trust in AI-assisted care.

VII. Conclusion

This study explores the integration of LLMs into personalized medication strategies for Parkinson's disease, demonstrating greater effectiveness, reliability, and interpretability compared to previous methods based on traditional machine learning and deep learning. Experiments on real-world datasets demonstrate that our LLM-driven system achieves over 1.4 points of MDS-UPDRS-III score reduction compared to human physicians. These findings highlight its clinical effectiveness and support its integration into medical practice.

Despite the encouraging outcomes achieved so far, further validation in larger and more diverse clinical settings is necessary to address challenges associated with patient heterogeneity. To this end, we propose two promising directions for future research.

First, the increasing availability of digital health technologies offers a promising opportunity to enhance the PD-LLM framework. Real-time patient data from wearable sensors, smartphone applications, and other monitoring tools enable continuous, objective tracking of symptom progression. For example, Woodrow et al. [51] have demonstrated that devices like the Parkinson's KinetiGraph (PKG) enhance PD management by monitoring bradykinesia and dyskinesia, enabling more precise treatment adjustments compared to traditional assessments. By integrating real-time data, our framework could dynamically update patient state with high-frequency, precise, and granular information, delivering more adaptive and personalized treatment recommendations. This would enhance the system's robustness and clinical utility.

Second, the complex, chronic nature of Parkinson's disease calls for multi-agent collaborative strategies. Multiple LLM-based agents, emulating specialists like neurologists, rehabilitation therapists, and nutritionists, could collaboratively optimize treatment plans across multiple dimensions of care. This coordinated approach would deliver comprehensive, adaptive, and effective disease management, improving scalability and efficacy. Moreover, such a multi-agent framework also opens new avenues for research and clinical applications in the comprehensive management of neurodegenerative diseases.

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